MATH317 · NUMERICAL METHODS

Midterm 471 Practice

20/10/2025 · 60 minutes · 30 marks

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01 Questions

Answer each question in the space below it. The full solution is in a separate file.

Q1 7 M

Let x = 0.54617 and y = 0.54601. Use four-digit arithmetic to approximate x − y and determine the absolute and relative errors using rounding of numbers.

Q2 8 M

Given the polynomial $P(x) = 3x^3 - 4x^2 - 12$.

a) Use Horner's algorithm to evaluate $P(-2)$.

b) Using the complete Horner's algorithm, find the Taylor expansion of $P(x)$ about the point $r = 3$.

Q3 7 M

Determine the single-precision machine representation of the decimal number (-45.21875).

Q4 8 M

Given $f(x) = x^3 - 9x + 1$.

a) Prove that $f(x)$ has exactly one real root α in the interval $[2, 4]$.

b) Find a root of $f(x)$ on $[2, 4]$ by using the bisection method. Terminate the iterations if they exceed nmax = 2 or if the error bound is less than 0.5×10^{-6}.